market_exposed_max

equity strategy | family: equity-max | status: research | derived from `market_exposed`

What this strategy does

market_exposed with bigger bets for more return: base unit 0.006, 6x leverage cap

Why it should work. Let net exposure float with the signal; empirically this smuggles in uncontrolled market beta.

Design notes (from the strategy's source file).

Max-size variants ("<name>_max") of the hand-built equity strategies.

Same signal, trim, tilt and neutralization as each base strategy; only the two dials that set how big and how levered the book is change (base_unit_fraction, gross-leverage cap = 6x). The multiplier for each was picked by sweeping 1.5x..12x the original base unit and taking the highest return subject to Sharpe >= 0.6 and max drawdown >= -40% (a worse drawdown is a fund-ending event, not a risk preference). The untouched 5%-of-ADV liquidity cap remains the real capacity constraint, and bites roughly four times as often at this sizing.

These are generated from the base strategy classes, so a change to a base strategy's weights flows through automatically.

parameter	value
base_unit_fraction	0.006
max_gross_leverage	6.0
liquidity_cap_frac	0.05
return_basis	raw
cap_mode	priority
exposure_cap	open_book
beta_overlay	0.0

Performance

metric	market exposed max	market (CRSP VW)
Annualized return	11.52%	9.94%
Annualized volatility	17.21%	19.35%
Sharpe (NAV-pct)	0.72	0.59
Sortino	1.04	
Max drawdown	-53.8%	-55.5%
Calmar	0.21	
Final NAV ($10M start)	$83,607,608	
Worst day	-8.00%	
% positive days	51.0%	
Sharpe 1996-2001	1.33	

Sharpe 2002-2007	1.13	
Sharpe 2008-2013	0.06	

Correlation of daily returns with the market: **0.57**.

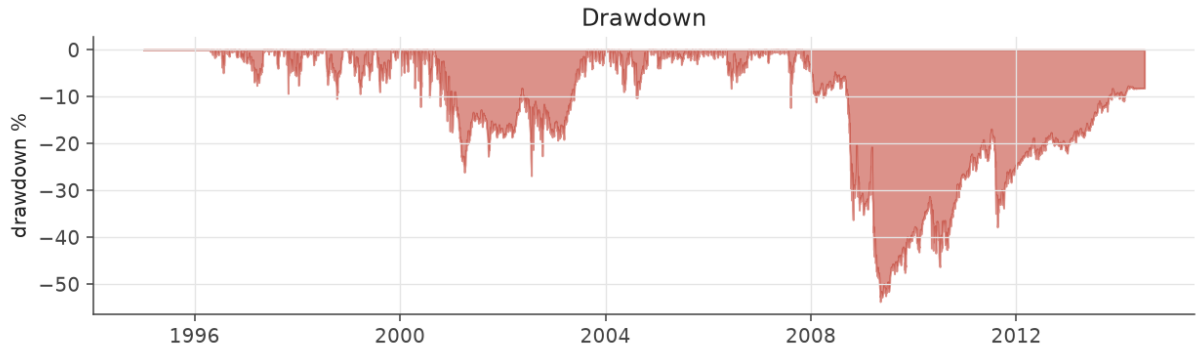
• best Sharpe among equity: `calm_rank_tilt` (1.81); this one: 0.72

• best return among equity: `market_boost_max` (0.15); this one: 0.115

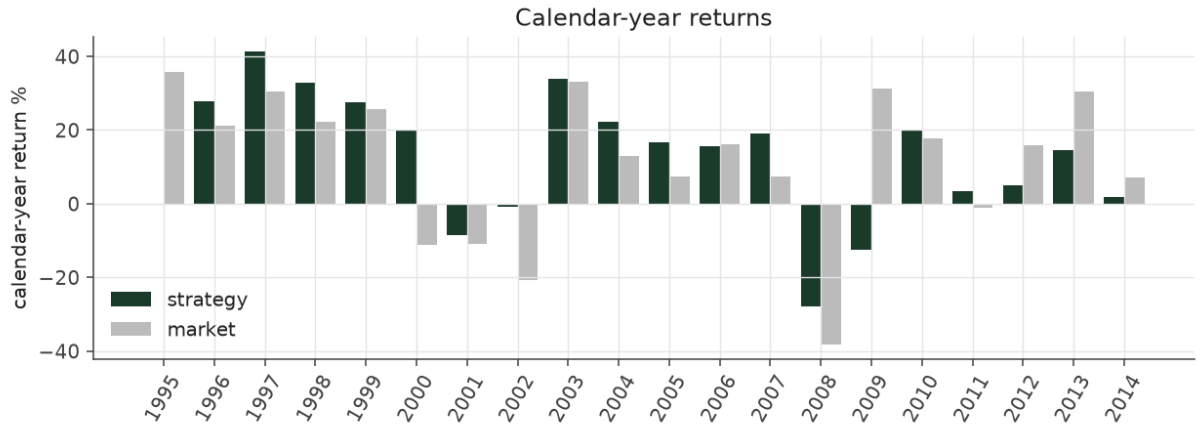
• best Calmar among equity: `calm_rank_tilt` (0.614); this one: 0.214



NAV (log scale)



Drawdown from running peak

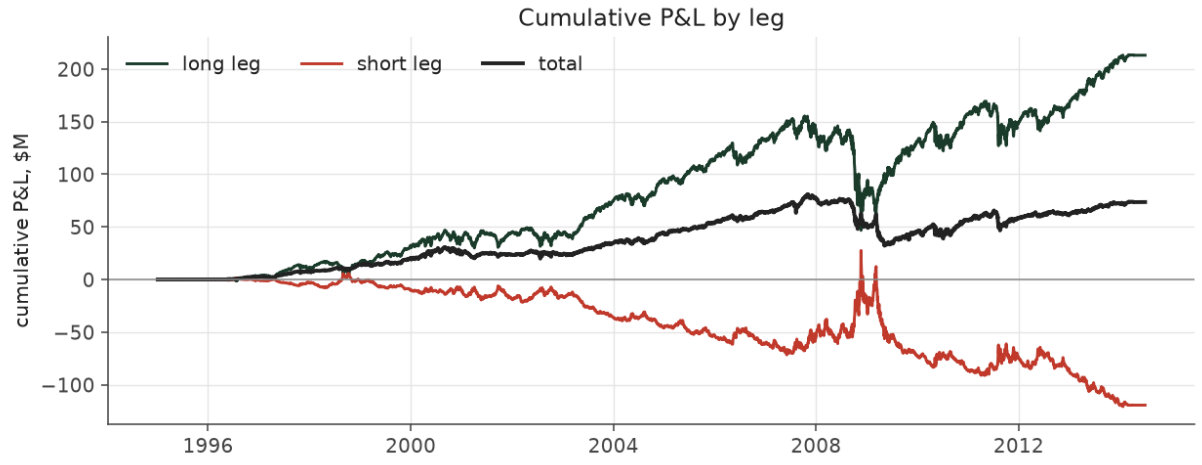


Calendar-year returns



Is the edge stable, or concentrated in a few years?

Where the return comes from



Long leg vs. short leg vs. total

	long leg	short leg
P&L ($M, before costs)	213.13	-119.36
P&L after costs ($M)	202.85	-129.25
of which common (average-event) component ($M)	80.70	-81.39

of which excess / PEAD component ($M)	27.98	41.62
excess minus costs ($M)	17.70	31.73
gross return per $ traded (bp)	345.33	-221.21
positions in the money (hit rate)	0.62	0.42
stand-alone Sharpe (leg only)	0.70	-0.44

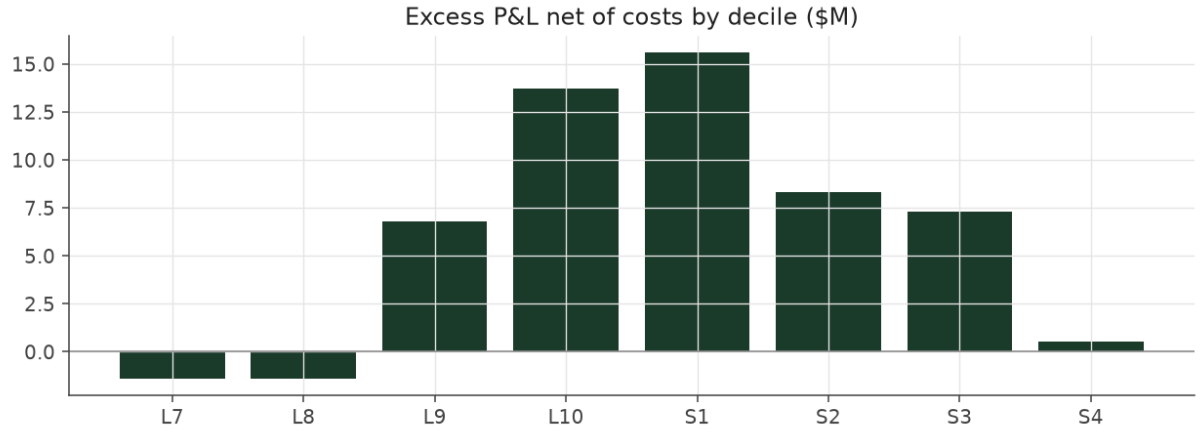
Daily P&L correlation between the two legs: **-0.93**.

• **The short leg loses money in absolute terms** (-119.4M before costs) while the long leg earns +213.1M. That is mostly a benchmark effect, not a signal failure: the average earnings-event stock beat the cap-weighted market during its holding window, which flatters the long leg and hurts the short leg by 80.7M / 81.4M. Stripping that common component, the surprise signal itself earned +28.0M on the long side and +41.6M on the short side.

• The two legs' daily P&L are -0.93 correlated: the short leg is the hedge that makes the long leg's variance disappear (stand-alone leg Sharpe 0.70 long / -0.44 short vs 0.72 combined).

By decile and side (signal-driven part only)

leg	decile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
long	7	6,527	417.61	9.33	-0.69	0.72	-1.42	-2.9%	-33.91
long	8	16,617	1,361.28	35.85	0.93	2.31	-1.38	-2.8%	-10.14
long	9	18,570	1,900.65	72.53	9.90	3.12	6.78	13.7%	35.68
long	10	20,136	2,492.30	85.14	17.84	4.13	13.71	27.7%	55.02
short	1	18,299	2,083.38	-46.96	19.47	3.88	15.60	31.6%	74.86
short	2	18,907	1,863.87	-48.19	11.70	3.38	8.32	16.8%	44.65
short	3	15,390	1,301.49	-29.54	9.67	2.38	7.29	14.7%	56.00
short	4	2,374	146.98	-4.56	0.78	0.26	0.53	1.1%	35.69



D10 and the short deciles carry the alpha; the middle of each leg earns little

By size quintile (1 = smallest)

size quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	14,061	528.31	-7.27	-0.21	1.85	-2.05	-4.2%	-38.87
2	22,253	1,721.50	13.34	18.59	4.47	14.12	28.6%	81.99

3	25,035	2,626.97	18.91	19.10	5.18	13.92	28.2%	53.01
4	27,193	3,241.88	18.08	16.09	4.90	11.18	22.6%	34.49
5	28,278	3,448.91	30.55	16.03	3.77	12.26	24.8%	35.55

By liquidity quintile (1 = least liquid)

liquidity quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	13,415	298.06	2.03	4.96	1.49	3.47	7.0%	116.36
2	22,552	1,606.91	7.47	14.23	4.82	9.41	19.0%	58.54
3	25,317	2,835.55	16.87	24.56	5.67	18.89	38.2%	66.62
4	27,157	3,397.79	13.94	10.68	4.76	5.92	12.0%	17.42
5	28,374	3,428.39	33.21	15.06	3.43	11.63	23.5%	33.93

By holding-window length

hold bucket	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
<=40d	41,781	3,901.53	-13.11	27.28	6.64	20.64	41.8%	52.91
41-60d	27,668	2,697.81	-2.50	15.18	5.20	9.98	20.2%	36.98
61-90d	37,860	4,270.22	64.61	16.01	6.67	9.34	18.9%	21.87
>90d	9,511	698.02	24.61	11.14	1.66	9.47	19.2%	135.69

By era

era	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1996-01	32,241	1,734.36	24.05	19.18	3.00	16.18	32.7%	93.32
2002-07	38,279	4,245.29	48.63	38.84	7.24	31.60	63.9%	74.44
2008-13	46,300	5,587.92	0.92	11.57	9.94	1.64	3.3%	2.94

By FF12 sector

ff12 sector	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
BusEq	21,153	1,230.94	17.81	4.34	2.31	2.04	4.1%	16.54
Chems	2,553	695.48	5.74	6.58	1.04	5.54	11.2%	79.62
Durbl	2,830	759.82	-1.46	1.25	1.16	0.09	0.2%	1.23
Enrgy	6,352	934.06	6.22	12.56	1.50	11.07	22.4%	118.48
Hlth	11,421	1,155.97	11.45	5.78	2.12	3.66	7.4%	31.67
Manuf	11,913	1,060.18	4.41	4.72	1.90	2.83	5.7%	26.65
Money	17,299	1,052.35	5.96	8.54	2.06	6.47	13.1%	61.53
NoDur	4,908	876.08	8.36	5.26	1.42	3.84	7.8%	43.83
Other	18,107	1,179.78	10.00	10.24	2.32	7.92	16.0%	67.09
Shops	11,583	1,114.80	9.28	7.06	1.99	5.08	10.3%	45.54

Telcm	3,961	799.00	-3.92	1.90	1.21	0.69	1.4%	8.63
Utils	4,740	709.11	-0.24	1.36	1.15	0.21	0.4%	3.00

By idiosyncratic-volatility tercile

vol tercile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
high vol	39,072	3,975.49	2.33	11.01	6.38	4.63	9.4%	11.65
low vol	38,994	3,691.64	38.90	32.77	6.98	25.79	52.2%	69.87
mid vol	36,964	3,759.99	31.67	25.27	6.53	18.74	37.9%	49.85

Short-borrow sensitivity (not modeled in the backtest)

borrow rate	fee M	ann. return	drag
0.0%	0.0	11.52%	0.00%
0.5%	5.3	11.14%	0.38%
1.0%	10.7	10.74%	0.78%
2.0%	21.3	9.85%	1.67%
5.0%	53.3	5.85%	5.67%

The engine charges the same transaction cost on shorts as on longs and no stock-borrow fee. The short leg is concentrated in small, illiquid names, where borrow is scarce and expensive -- read the table as a floor on the cost of running the short leg.